Certificate-Only Causal Discovery: Monitoring-Valid Edges and Predictive-Fit Orientation

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Abstract

Constraint-based causal discovery deletes an edge when a conditional-independence test fails to reject, so a lack of power is reported as a conclusion, and its output cannot be monitored as data arrive: an analyst who reruns it at each sample size and keeps any edge ever reported accumulates false edges. We study certificate-only discovery, which asserts a feature of the graph only by rejecting a null that is true whenever the feature is absent. The adjacency certificate is the minimum over conditioning sets of sequential tests, either a Bayes-factor test of a partial regression coefficient (exact under Gaussian errors) or a nonparametric sequential generalised covariance test that allows nonlinear mechanisms; it bounds the probability that a false edge is ever certified, uniformly over time and without faithfulness, exactly or asymptotically. The orientation certificate compares sequentially fitted forecasters for the two factorisations of a pair and certifies a direction only when two model classes agree. In 150 random graphs monitored from $n = 100$ to **6400**, CERT-CD reported a false edge in **2.7%** of graphs under Laplace noise, against **27%** to **91%** for PC, GES and DirectLiNGAM, at the price of leaving most non-adjacent pairs undecided and a low recall at small samples. With nonlinear mechanisms the linear test reported a false edge in about half of the graphs and the nonparametric test in **1** to **2** of **60**. On the Tübingen cause-effect pairs, under a protocol fixed before the run, the orientation certificate committed on **12** of **80** scorable pairs and was right on **11**. On two physical devices with known graphs (Causal Chambers), every method including the first version of CERT-CD reported edges the graph does not contain; after a redesign made on these data, the nonparametric certificate reported none outside the graph and the pairs the graph cannot explain in **9** of **10** and **10** of **10** subsamples, with low recall. Both certificates fail outside their models, and we report where.

Keywords: causal discovery, anytime-valid inference, e-values, sequential testing, additive noise models, family-wise error control

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1 Introduction

A causal discovery algorithm returns a graph, and a practitioner reading it wants to know which of its features the data support. Constraint-based methods such as PC and FCI (Colombo, Maathuis, Kalisch, & Richardson, 2012; Spirtes, Glymour, & Scheines, 2000; Spirtes, Meek, & Richardson, 1995) start from the complete graph and delete an edge whenever a conditional-independence test fails to reject. Failure to reject is compatible with independence, with a dependence too weak to see at the current sample size, and with a test that had little power; the deletion happens regardless. Two consequences follow. Finite-sample correctness requires faithfulness or a quantitative strengthening of it (Robins, Scheines, Spirtes, & Wasserman, 2003; Uhler, Raskutti, Bühlmann, & Yu, 2013), because only such an assumption turns "the test did not reject" into "the variables are independent". And the output is two-valued where the evidence is three-valued: a missing edge means either "the data indicate no edge" or "nothing was established about this pair".

There is a third consequence that matters in practice. Data accumulate, and an analyst who inspects the estimate as they do, and keeps any edge that has ever appeared, is performing optional stopping. A fixed-sample guarantee does not cover this. Section 4.2 measures the effect on eight-node graphs: when PC, GES and DirectLiNGAM are rerun at each sample size from 100 to 6400, between a quarter and nine tenths of the graphs have at least one false edge in the union of the estimates.

This paper studies the opposite discipline, which we call *certificate-only* discovery: assert a feature of the graph only by rejecting a null hypothesis that is true whenever the feature is absent, and never assert anything by failing to reject. An algorithm that obeys the rule starts from the empty graph and adds; it cannot report an edge as absent, only as undecided; and what it controls is the probability of asserting something false. If each test is an anytime-valid process, the guarantee holds at every sample size at once and survives continuous monitoring.

Contributions.

1. *Adjacency certificates under monitoring.* In a structural equation model, d-separation implies that the conditional covariance of a pair given a separating set vanishes, so adjacency has a certifying null that needs no faithfulness. The minimum over conditioning sets of per-set sequential tests bounds the probability that a false edge is *ever* certified (Section 3.4). We give two per-set tests: a Bayes factor for a linear model, exact for Gaussian errors, and a nonparametric sequential generalised covariance test, asymptotic, that allows nonlinear mechanisms (Section 3.4.2). We measure how far from nominal they are off-model (Sections 4.6 and 4.5).

2. *An orientation certificate based on predictive fit.* For a pair of variables we fit forecasters for the factorisations $x \to y$ and $y \to x$ sequentially, and certify a direction when the running mean of the log-score difference has a confidence sequence that excludes zero under two model classes at once (Section 3.5). The claim is about predictive fit and is valid whether or not either model is correct; it identifies the causal direction under additive-noise models.
3. *Evidence under pre-specified protocols.* We compare against PC, GES and DirectLiNGAM under monitoring (Section 4.2), evaluate the orientation certificate on simulated pairs and on the Tübingen cause-effect pairs with a protocol committed before the run (Sections 4.8 and 4.9), evaluate it on physical devices with a known graph (Section 4.10), study scaling with graph size, sample size and the constants of the algorithm (Sections 4.3 and 4.4), and report the regimes where the certificates fail (Sections 4.8 and 4.11).

Scope.

The method targets continuous, observational data from a causally sufficient system whose mechanisms are linear or additive-noise, at sample sizes of a few hundred or more. We chose the benchmarks to match this domain and fixed the selection criteria before running; data outside it are reported as a limitation, not dropped.

Outline.

Section 2 places the work in the literature. Section 3 develops the certificates and the algorithm. Section 4 reports the experiments and Section 5 discusses limitations. Proofs and further results are in the appendices.

2 Related work

Error control in constraint-based discovery.

PC-p (Strobl, Spirtes, & Visweswaran, 2019) bounds an edge's p-value by the maximum p-value over conditioning sets and controls the false discovery rate of edge and orientation decisions with Benjamini–Yekutieli; it is finite-sample but valid at a fixed sample size, and its orientation rules operate within the Markov equivalence class. The conservative PC algorithm of Ramsey, Spirtes, and Zhang (2006) marks ambiguous triples instead of resolving them. Faithfulness-based guarantees break down when parameters are near a cancellation (Robins et al., 2003; Uhler et al., 2013; Zhang & Spirtes, 2008). ϵ-CUT (Shaska & Mitra, 2025) gives one-sided false-edge control without faithfulness at each fixed sample size; Csillag, Struchiner, and Goedert (2025) calibrate batched Fisher-z statistics into per-test e-values inside PC. The certificate-only discipline here is the same one-sided logic made time-uniform and extended to orientation.

Anytime-valid inference.

Ville's inequality (Ville, 1939) underlies confidence sequences (Darling & Robbins, 1967; Howard, Ramdas, McAuliffe, & Sekhon, 2021; Robbins, 1970) and the e-value literature (Grünwald, de Heide, & Koolen, 2024; Ramdas, Grünwald, Vovk, & Shafer,

2023; Vovk & Wang, 2021). Asymptotic confidence sequences (Waudby-Smith, Arbour, Sinha, Kennedy, & Ramdas, 2024) and anytime-valid linear models (Lindon, Ham, Tingley, & Bojinov, 2026) extend the approach to regression coefficients under weak moment conditions. Conditional-independence testing is intrinsically hard: no test with uniform level over all conditional-independence nulls has power against any alternative (Shah & Peters, 2020). Sequential tests therefore either assume a known conditional law (Grünwald, Henzi, & Lardy, 2024; Shaer, Maman, & Romano, 2023) or restrict the null, as we do by testing a partial regression coefficient.

Orientation.

Linear non-Gaussian models identify the full graph (Shimizu, Hoyer, Hyvärinen, & Kerminen, 2006) and DirectLiNGAM (Shimizu et al., 2011) and the pairwise likelihood ratios of Hyvärinen and Smith (2013) estimate it; Shimizu and Kano (2008) test direction using non-normality. Additive-noise models (Hoyer, Janzing, Mooij, Peters, & Schölkopf, 2009; Peters & Bühlmann, 2014) identify direction for nonlinear mechanisms, and location-scale models extend this to heteroscedastic noise (Immer et al., 2023; Xu, Mian, Marx, & Vreeken, 2022). Compression- and quantile-based scores (Marx & Vreeken, 2017; Tagasovska, Vatter, & Chavez-Demoulin, 2020) perform well on the Tübingen pairs (Mooij, Peters, Janzing, Zscheischler, & Schölkopf, 2016). Linear methods fit those pairs poorly (Schkoda & Drton, 2025), and a test that is valid against a misspecified model does not control direction error. Methods that abstain exist: Prakash, Xia, and Erosheva (2024) and Ding and Zhang (2025) return an inconclusive outcome, Prakash, Xia, and Erosheva (2026) gate on a linearity check, and Uehara (2026) attach per-edge resolved or impossible codes in an interactive protocol. None of these gives a time-uniform error statement; ours is a statement about predictive fit that holds without either model being correct (Section 3.5), a comparison in the spirit of Vuong (1989) and of sequential forecaster comparison (Choe & Ramdas, 2024).

3 Method

3.1 Overview

CERT-CD has two certificates and one rule for combining them. The *adjacency certificate* for a pair is a sequential test of the hypothesis that some admissible conditioning set makes the partial regression coefficient zero; it fires when the evidence against that hypothesis, minimised over conditioning sets, is large. The *orientation certificate* for a pair compares two sequentially fitted forecasters of the joint law and fires when one factorisation predicts strictly better. The combining rule is that the graph is built only from fired certificates, starting from the empty graph, so that the output of a run is a set of edges and arrowheads each of which carries a stated bound on the probability that it is ever wrong. Table 1 collects the notation.

Table 1 Notation.

$\mathcal{G}^\star$, d	true DAG and number of variables
$\mathcal{S}_k(i,j)$	admissible conditioning sets of size at most k
$E_t^{(S)}(i,j)$	Bayes-factor process for $\beta_{i\leftarrow j\cdot S}=0$ after t rows
$A_t(i,j)$	adjacency evidence, minimum over $S\in\mathcal{S}_k(i,j)$
g, $G=gS_{jj}$	prior variance factor and its data-scaled version
$\alpha=\alpha_A+\alpha_D$	total error budget split between adjacency and direction
λ_A, λ_D	fixed thresholds for the two certificate families
d_s	log-score difference of the two forecasters at observation s
m_0	warm-up length, excluded from every monitored process

3.2 Setting and assumptions

Let $\mathcal{G}^\star$ be a directed acyclic graph on $V=\{1,\dots,d\}$ and let $X=(X_1,\dots,X_d)$ follow a linear structural equation model $X=B^\top X+e$, where B is supported on the edges of $\mathcal{G}^\star$ and the errors $e_1,\dots,e_d$ are mutually independent with finite fourth moments. Observations $o_1,o_2,\dots$ are i.i.d. draws of X and arrive in batches; $\mathcal{F}_t$ is the σ-field of the first t observations. We write $\mathrm{pa}(i)$ and $\mathrm{de}(i)$ for the parents and descendants of i and $\mathcal{S}_k(i,j)=\{S\subseteq V\setminus\{i,j\}:|S|\le k\}$ for the admissible conditioning sets, a family fixed before any data are seen.

Assumption 1 (Separator size) Every non-adjacent pair of $\mathcal{G}^\star$ has a d-separating set of size at most k.

Assumption 1 is structural. It holds whenever k is at least the maximum in-degree of $\mathcal{G}^\star$, because for non-adjacent i,j either $\mathrm{pa}(i)$ or $\mathrm{pa}(j)$ d-separates them; $k=d-2$ makes it vacuous at an exponential computational cost.

For a pair (i,j) and a set S let $\beta_{i\leftarrow j\cdot S}$ denote the coefficient of X_j in the population regression of X_i on $(1,X_j,X_S)$.

Lemma 1 *If S d-separates i and j in $\mathcal{G}^\star$, then $\beta_{i\leftarrow j\cdot S}=0$.*

Proof sketch In a linear structural equation model with independent errors the covariance of X is $\Sigma=(I-B)^{-\top}\Omega(I-B)^{-1}$ with Ω diagonal, and the partial covariance of X_i and X_j given X_S is a rational function of the entries of Σ. It depends on the error laws only through Ω. For Gaussian errors, d-separation of i and j by S implies conditional independence by the global Markov property and hence a zero partial covariance (Spirtes et al., 2000); the partial covariance is the same function of (B,Ω) for any other error laws with the same variances, so it vanishes for them as well. The population regression coefficient $\beta_{i\leftarrow j\cdot S}$ is proportional to the partial covariance. □

This is a moment condition, weaker than conditional independence, and it does not require Gaussian errors. Faithfulness is the converse and is not assumed.

3.3 Certificates

Definition 1 Let ϕ be a feature of $\mathcal{G}^\star$. A *certifying null* for ϕ is a hypothesis H_ϕ that holds whenever ϕ is false. A process $(E_t)_{t\geq 0}$ is *Ville-valid* for H if $E_t \geq 0$ and $\mathbb{P}(\exists t : E_t \geq \lambda) \leq 1/\lambda$ for every $\lambda \geq 1$ and every $P \in H$. A *certificate for ϕ at level α* is the event that a Ville-valid process for H_ϕ reaches $1/\alpha$.

Proposition 2 (Soundness) *If ϕ is false, the probability that it is ever certified at level α is at most α.*

The proof is the definition: if ϕ is false then $P \in H_\phi$. The implication runs one way. H_ϕ may also hold when ϕ is true, and the certificate then never fires; this is what buys validity without faithfulness and what costs power. Three consequences follow. There is no certifying null for "i and j are non-adjacent", so the algorithm cannot delete and must start from the empty graph. Its output is three-valued (certified adjacent, certified with a direction, undecided). And power is descriptive: nothing guarantees that a true edge is ever certified.

Ville-valid is weaker than *e-process*, which requires $\mathbb{E}[E_\tau] \leq 1$ at every stopping time of a single filtration. All our bounds use only the event that a process ever reaches a level, so they hold at every data-dependent stopping time of the run, but we do not claim the stronger property. Section 3.4.1 shows why.

3.4 The adjacency certificate

For a pair (i, j) consider

$$H_{ij}^{\text{sep}} : \quad \exists S \in \mathcal{S}_k(i,j) \;\text{ with }\; \beta_{i \leftarrow j \cdot S} = 0. \tag{1}$$

By Lemma 1 and Assumption 1, H_{ij}^{sep} holds for every non-adjacent pair, so it is a certifying null for adjacency. Let $E_t^{(S)}$ be a Ville-valid process for the point null $\beta_{i \leftarrow j \cdot S} = 0$ and define

$$A_t(i,j) = \min_{S \in \mathcal{S}_k(i,j)} E_t^{(S)}(i,j). \tag{2}$$

Proposition 3 *Let $\lambda \geq 1$ and let (i, j) be non-adjacent. Then $\mathbb{P}(\exists t : A_t(i,j) \geq \lambda) \leq 1/\lambda$.*

Proof Choose $S^\star \in \mathcal{S}_k(i,j)$ with $\beta_{i \leftarrow j \cdot S^\star} = 0$. Since $A_t \leq E_t^{(S^\star)}$ pointwise, the event $\{\exists t : A_t \geq \lambda\}$ is contained in $\{\exists t : E_t^{(S^\star)} \geq \lambda\}$, which has probability at most $1/\lambda$. □

That the minimum of e-values is an e-value for a union null is standard (Vovk & Wang, 2021); the proof above needs only the inclusion of events, so the processes $E^{(S)}$ may live in different filtrations. Faithfulness enters only through power: adjacency-faithfulness makes every $E^{(S)}$ grow for a genuinely adjacent pair, and an instance where it fails leaves the pair undecided instead of producing a wrong edge.

3.4.1 The per-set primitive

For S with $|S| = q - 1$ write the regression $X_i = \alpha^\top(1, X_S) + \beta X_j + \varepsilon$ and let S_{ii}, S_{ij}, S_{jj} be the cross-products of the residuals of X_i and X_j after regression on $(1, X_S)$, computed from the running Gram matrix of the centred data, with $p = |S|+1$. With a right-Haar prior on the nuisance parameters and $\beta \mid \sigma \sim N(0, \sigma^2 g)$ for a fixed $g > 0$, the Bayes factor is

$$E_n = (1+G)^{-1/2}\big(1 - \kappa\, \hat{r}_n^2\big)^{-(n-p)/2}, \qquad G = g\, S_{jj}, \quad \kappa = \tfrac{G}{1+G}, \quad \hat{r}_n^2 = \tfrac{S_{ij}^2}{S_{ii} S_{jj}}. \tag{3}$$

It depends on the data only through Schur complements of the Gram matrix, so a single $O(d^2)$ statistic serves every hypothesis (i, j, S). The prior scale g must be fixed before the data: a g that adapts to the running sample size destroys the martingale property. We fix it in advance and use a warm-up prefix, discarded from every process, only to standardise the variables.

Proposition 4 *Suppose $\varepsilon \sim N(0, \sigma^2)$ is independent of (X_j, X_S) and $\beta = 0$. Then (E_n) is a non-negative martingale with mean one in the filtration generated by the sequence of maximal invariants of the nuisance group (translations in the span of $(1, X_S)$ and rescaling of X_i), and therefore $\mathbb{P}(\exists n : E_n \geq \lambda) \leq 1/\lambda$ for every $\lambda \geq 1$.*

The proof is the right-Haar argument of Hendriksen, de Heide, and Grünwald (2021) and Grünwald et al. (2024) and is given in Appendix A. The regressors may have any distribution; Gaussianity is required of the regression error ε, which for $S \supseteq \mathrm{pa}(i)$ with no descendant of i is the structural error e_i.

Remark 1 (The raw-observation filtration) (E_n) is *not* a martingale in the filtration generated by the raw observations. With $S = \emptyset$, $g = 0.25$ and $X_{1:2} = Y_{1:2} = (-1, 1)$, Eq. (3) gives $E_2 = 1$ while $\mathbb{E}[E_3 \mid \mathcal{F}_2] = 1.083 > 1$ under an i.i.d. standard Gaussian null (numerical integration). Amenability of the nuisance group does not repair this. The guarantee used here, a bound on the probability of ever crossing, does not need the raw filtration, which is why Proposition 3 was stated for events.

Remark 2 (Role averaging) Eq. (3) treats X_i as the response, and the roles of i and j are interchangeable under the null. The implementation averages the two role-swapped Bayes factors to remove the arbitrary choice. A crossing of the average implies a crossing of one of the two, each of probability at most $1/\lambda$, so the guaranteed bound for the averaged process is $2/\lambda$.

Remark 3 (Beyond Gaussian errors) When the regression error is not Gaussian, or its variance depends on the regressors, Eq. (3) is the $d = 1$ case of the model-based test of Lindon et al. (2026), whose guarantee under the moment null $E[W\varepsilon] = 0$ is asymptotic and holds when the model variance equals the heteroscedasticity-robust variance. A robust sandwich version exists (Lindon et al., 2026, Theorem 5.4 of the arXiv version). Finite-sample validity over all finite-variance laws is impossible without tail control (Shah & Peters, 2020). We

therefore state the adjacency guarantee as *exact for Gaussian regression errors and asymptotic otherwise*, and Section 4.6 measures the finite-sample departure under heavy tails and heteroscedasticity for both versions.

Remark 4 (One model for both certificates) Both certificates live in the linear structural equation model with independent errors. Adjacency uses only second moments, through Lemma 1; orientation uses the shape of the error laws and, in the additive-noise extension, nonlinear mechanisms, for identifiability only. The validity statement of the orientation certificate (Section 3.5) does not depend on either error law.

3.4.2 A nonparametric primitive

The Bayes factor above needs the regression of X_i on (X_j, X_S) to be linear with a constant-variance error. For the dependences that structural equations with nonlinear mechanisms produce, this is too much to ask: if S contains a mediator between i and j whose effect is nonlinear, a linear regression leaves residual dependence, and the pair is certified although it is non-adjacent (Section 4.5). We therefore give a second primitive, a sequential version of the generalised covariance measure of Shah and Peters (2020), in the spirit of the anytime-valid conditional independence test without the model-X assumption of Waudby-Smith, Kennedy, and Ramdas (2023). Write $r_i^S = X_i - \mathbb{E}[X_i \mid X_S]$. If S d-separates i and j then $X_i \perp\!\!\!\perp X_j \mid X_S$ and

$$\mathbb{E}\big[r_i^S \, r_j^S\big] = 0, \tag{4}$$

with no assumption on the form of the conditional means. We use three versions of the product, $r_i^S r_j^S$ and the two products in which one factor is the residual of the square, $r_i^S \, (X_j^2 - \mathbb{E}[X_j^2 \mid X_S])$ and its transpose; each is zero under the separation null, and the squares let the test see dependences that are symmetric, such as $X_j = X_i^2 + \epsilon$, which the plain covariance misses.

Construction.

Let $\hat{f}^S_{v,s}$ be an estimate of $\mathbb{E}[X_v \mid X_S]$ computed from the rows before row s only, for $v \in \{i, j\}$ (and for the squares). The scored product is $z_s = (X_{i,s} - \hat{f}^S_{i,s}(X_{S,s}))(X_{j,s} - \hat{f}^S_{j,s}(X_{S,s}))$. Its running mean has the asymptotic confidence sequence of Waudby-Smith et al. (2024, Theorem 2.2), as in Eq. (6). The set S is *rejected* at the first time at which the sequence for any of the three products excludes zero (level $\alpha_{\text{pair}}/3$ each), and the pair is certified adjacent once *every* $S \in \mathcal{S}_k(i, j)$ has been rejected.

Proposition 5 *Let (i, j) be non-adjacent in $\mathcal{G}^\star$ and let $S^\star \in \mathcal{S}_k(i, j)$ separate them. Suppose the conditional-mean estimates are predictable and the moment conditions of Waudby-Smith et al. (2024) hold for z_s, and that the product of the estimation errors, $\mathbb{E}[(f - \hat{f}_i)(g - \hat{f}_j)]$, has a running average that is $o_P(t^{-1/2})$. Then the probability that $S^\star$ is ever rejected is at most α_{pair}, asymptotically, and so is the probability that the pair is ever certified.*

Proof sketch Under (4) the conditional mean of z_s given the past is the product of the two estimation errors, so z_s has a martingale-difference part with mean zero and a predictable remainder; the remainder is negligible by the product-rate condition, and the confidence sequence covers zero uniformly in t with asymptotic probability at least $1 - \alpha_{\text{pair}}/3$ per product. A union bound over the three products gives α_{pair}. The pair is certified only if $S^\star$ is rejected, so its probability is smaller. □

The guarantee is asymptotic, and it is conditional on the estimation error being small: no sequential test of conditional independence is valid uniformly over all laws (Shah & Peters, 2020). In exchange it covers nonlinear conditional means, which the linear primitive does not.

Implementation.

The regression of each X_v and X_v^2 on X_S is a ridge-penalised additive cubic spline (three knots at the quartiles of the warm-up, inputs clipped at the 1% and 99% warm-up quantiles) with the products of pairs of variables added when $|S| = 2$; it is refit on all rows seen so far when the number of rows has grown by a factor 1.25, and the residuals of the current fit are used for scoring the rows that follow. One multi-output solve per S serves all variables, so a pass costs $O\big(\sum_{|S|\le k} n\, p_S^2\big)$ for the refits and $O(d^2)$ per set per batch for scoring. For $d = 8$ and $k = 2$ a run on 3200 rows takes about 0.3 seconds, and for $d = 20$ and $k = 3$ (about 1350 sets) about 25 seconds on 4000 rows.

3.5 The orientation certificate

Fix a pair (x, y). For each direction we fit a forecaster of the joint law on the past and score it on the next observation. Let $p_s^{x\to y}$ be the density of the forecaster for the factorisation $p(x)\,p(y \mid x)$ fitted on observations $1, \dots, s-1$ and $p_s^{y\to x}$ the mirror forecaster, and set

$$d_s = \log p_s^{x\to y}(x_s, y_s) - \log p_s^{y\to x}(x_s, y_s). \tag{5}$$

Both forecasters are $\mathcal{F}_{s-1}$-measurable, and both are densities on the same space, so d_s is a legitimate score difference whether or not either model is correct. Let $\bar\mu_t = t^{-1}\sum_{s\le t}\mathbb{E}[d_s \mid \mathcal{F}_{s-1}]$ and let $\hat\mu_t$ and $\hat\sigma_t$ be the running mean and standard deviation of (d_s). The confidence sequence of Waudby-Smith et al. (2024, Theorem 2.2) is

$$\hat\mu_t \pm \hat\sigma_t \sqrt{\frac{2(t\rho^2+1)}{t^2\rho^2}\log\frac{\sqrt{t\rho^2+1}}{\alpha}}, \tag{6}$$

with ρ tuned to a reference time $t^* = 200$. We certify $x \to y$ at the first t at which the lower end exceeds zero and $y \to x$ when the upper end falls below zero; both are decided only after 60 scored observations.

Proposition 6 *Let $\mathcal{F}_{s-1}$ be the past, let the forecasters be $\mathcal{F}_{s-1}$-measurable, and suppose the martingale differences $d_s - \mathbb{E}[d_s \mid \mathcal{F}_{s-1}]$ satisfy the moment conditions of Waudby-Smith et al. (2024, Theorem 2.2). Then (6) is an asymptotic confidence sequence for $\bar\mu_t$ at level*

$1-\alpha$. *In particular, if* $y \to x$ *predicts at least as well as* $x \to y$ *on average* ($\bar{\mu}_t \le 0$ *for all* t), *the probability of ever certifying* $x \to y$ *is at most* α *asymptotically.*

What is certified is therefore that one factorisation has a strictly better expected log score, and this is valid without either forecaster being correct. It corresponds to the causal direction under the identifiability results for additive-noise models (Hoyer et al., 2009; Peters & Bühlmann, 2014) and for linear non-Gaussian models (Shimizu et al., 2006): if the data come from $x \to y$, the true class is contained in the forecaster family and the estimates are consistent, the expected score of the correct factorisation exceeds that of the reversed one by a positive Kullback–Leibler gap, because for generic mechanisms or non-Gaussian errors the reversed factorisation cannot represent the joint law. Under confounding or selection the better predictor need not be the cause, and the certificate then says nothing about causation. The conclusion of Proposition 6 is asymptotic, and Section 4.8 reports its finite-sample behaviour.

Forecasters.

Both use a generalised-Gaussian density $g(u) \propto \exp(-|u/s|^\kappa)$ with shape $\kappa \in [0.9, 8]$ fitted by maximum likelihood, a generalised-Gaussian marginal for the cause, and a cubic truncated-power spline mean with four quantile knots, set from the warm-up prefix, for the effect. The class `gg` has constant scale; the class `ggh` lets the log-scale be quadratic in the cause, which covers heteroscedastic noise (Immer et al., 2023; Xu et al., 2022). The *agreement* certificate reports a direction only when both classes certify the same one. Each class is Ville-bounded by the same argument, so agreement cannot err more often than either alone; its purpose is protection against model-class mistakes: the constant-scale class alone certifies the wrong direction on every heteroscedastic Gaussian pair it decides in our simulations, and the location-scale class alone is too flexible to resolve linear non-Gaussian pairs (Section 4.8). Preprocessing is fixed in advance: tied values are dequantised, the order is randomly permuted, and $\max(100, \lceil 0.2n \rceil)$ observations form a warm-up prefix used for standardisation and knots.

Fitting.

Both classes standardise x and y on the warm-up and set four knots at the $20, 40, 60, 80\%$ quantiles of the warm-up. The conditional density of the effect given the cause is fitted by maximising the generalised Gaussian likelihood with the mean a cubic truncated-power spline of the cause, by iteratively reweighted least squares with a continuation over the shape grid, so that the Gaussian shape $\kappa = 2$ and the Laplace shape $\kappa = 1$ are both reachable. In class `ggh` the scale is $\exp(\gamma_0 + \gamma_1 x + \gamma_2 x^2)$, re-estimated by two alternating rounds of weighted refitting. The forecasters are refit on all rows before each batch of 50 scored rows, and each scored row contributes one term d_s. A direction is reported at the first batch after 60 scored rows at which the confidence sequence excludes zero, and the verdict is not revised.

Proof of Proposition 6 The increments d_s are $\mathcal{F}_s$-measurable and the forecasters are $\mathcal{F}_{s-1}$-measurable, so $d_s - \mathbb{E}[d_s \mid \mathcal{F}_{s-1}]$ is a martingale difference sequence. Under the moment

Algorithm 1 Relative-fit orientation of a pair

Require: $(x_s, y_s)_{s \le n}$, level α, model classes $\{\texttt{gg}, \texttt{ggh}\}$

dequantise ties; permute rows at random; standardise and place knots using the first $m_0 = \max(100, \lceil 0.2n \rceil)$ rows
if $n - m_0 < 100$ **then return** UNSCORABLE
end if
for each class c **do**
  for each batch $s \in \{m_0, m_0 + 50, \dots\}$ **do**
    fit $p^{x \to y}, p^{y \to x}$ in class c on rows $1..s-1$; score the next 50 rows, append the d_s
    if at least 60 scored and no verdict yet **then** compute $\hat{\mu}_t \pm r_t$ from Eq. (6); if the lower end > 0 set verdict $x \to y$, if the upper end < 0 set verdict $y \to x$
    end if
  end for
end for
return the common verdict if both classes agree, else UNDECIDED

conditions, Theorem 2.2 of Waudby-Smith et al. (2024) gives a confidence sequence for the average conditional mean $\bar{\mu}_t$ that holds asymptotically, uniformly in t. If $\bar{\mu}_t \le 0$ for all t then crossing of the lower end above zero implies that the interval fails to cover $\bar{\mu}_t$ at some t. □

What the claim does and does not say.

The proposition is a statement about which forecaster scores better on the sequence of fitted models, not about the data-generating process. If neither class contains the truth, the better-scoring direction can still be the wrong causal one; the certificate is then valid for the statement it makes and misleading as a causal claim. Its causal reading needs identifiability, and then only if the fitted classes approximate the truth well enough for the gap in expected log score to have the right sign. The agreement rule reduces, but does not remove, this risk, and Section 4.8 measures it in simulation.

3.6 The algorithm and its guarantee

CERT-CD maintains a three-valued state for every pair. It streams data in batches, updates the running Gram matrix, recomputes $A_t(i, j)$ for every undecided pair, and certifies adjacency when $A_t(i, j)$ crosses its threshold. The total budget is split as $\alpha = \alpha_A + \alpha_D$, with α_A spread over the $\binom{d}{2}$ adjacency nulls and α_D over the direction nulls, and the thresholds are $\lambda_A = \binom{d}{2}/\alpha_A$ and $\lambda_D = d(d-1)/\alpha_D$, fixed before the data arrive. Which pairs are examined does not enter the bound.

Theorem 7 (Adjacency) *Under Assumption 1, linear structural equations with independent errors, and Gaussian regression errors in (3) for every examined (i, j, S), the probability that CERT-CD ever certifies an edge between a non-adjacent pair is at most $2\alpha_A$, uniformly over time and at every data-dependent stopping time of the run.*

Algorithm 2 CERT-CD (streaming)

Require: d, level α, split $\alpha_A = \alpha_D = \alpha/2$, order k, prior scale, batches $B_1, B_2, \ldots$

fix $\lambda_A = \binom{d}{2}/\alpha_A$, $\lambda_D = d(d-1)/\alpha_D$; set all pairs to UNDECIDED
standardise with mean and scale of the warm-up rows; discard them
for each batch B_t **do**
    update the Gram matrix with the standardised rows of B_t
    for each undecided pair (i, j) **do**
        $A \leftarrow \min_{S \in \mathcal{S}_k(i,j)}$ role-averaged $E^{(S)}$ from Eq. (3)
        record $\bar{A}(i, j) \leftarrow \max(\bar{A}(i, j), A)$
        if $\bar{A}(i, j) \geq \lambda_A$ **then** mark (i, j) ADJACENT; freeze $Z_{ij} \leftarrow$ certified neighbours of i and j; open two direction processes
        end if
    end for
    for each open direction process $(a \to b)$ **do**
        extend the numerator on B_t with a forecaster fitted on rows before B_t since opening
        if running maximum of log of the ratio $\geq \log \lambda_D$ **then** mark $a \to b$ certified
        end if
    end for
    output the current marked graph (certified edges, certified arrowheads, all else undecided)
end for

Proof By Remark 2 each pair's averaged minimum has crossing probability at most $2/\lambda_A$ by Proposition 3. A union bound over at most $\binom{d}{2}$ non-adjacent pairs gives $\binom{d}{2} \cdot 2/\lambda_A = 2\alpha_A$. □

Choosing $\lambda_A = 2\binom{d}{2}/\alpha_A$ gives α_A exactly; the experiments use the shipped threshold, so the guaranteed bound there is $2\alpha_A$. Under non-Gaussian errors the bound holds asymptotically (Remark 3). With the nonparametric primitive of Section 3.4.2 each pair has the level $\alpha_A/\binom{d}{2}$ and no factor 2 arises, so the probability that a false edge is ever certified is at most α_A asymptotically, without linearity or Gaussian errors. The orientation layer inside the multivariate algorithm uses a linear generalised-Gaussian universal-inference certificate whose validity is conditional on the model describing each pair given a block of conditioning variables frozen when the edge is certified (Appendix B); the relative-fit certificate of Section 3.5 is bivariate and is evaluated on pairs. An extension that conditions the relative-fit forecasters on the frozen block is possible but not evaluated here.

3.7 Implementation of the streaming algorithm

Algorithm 2 gives the streaming procedure. The sufficient statistic is the Gram matrix of the standardised rows, updated in $O(bd^2)$ per batch of b rows. Standardisation constants are computed once from the warm-up prefix of m_0 rows, which are then discarded; after that, no quantity used by a monitored process depends on the future.

Complexity.

Each pair has $\sum_{s \le k} \binom{d-2}{s}$ conditioning sets, so one pass over a batch costs $O\big(d^2\, d^k\, k^3\big)$ for the Schur complements, independently of n, and the adjacency state is $O(d^2)$. The direction processes keep the rows since the pair was certified and refit on them, so their cost grows with the time since certification. In our implementation the adjacency certificates alone take about 0.1 second for a run on 3200 rows at $d = 8$, and the full algorithm with orientation about 53 seconds, so the orientation processes dominate the cost (Sections 4.3 and 4.4). Certified pairs are skipped, and the evaluation of A_t stops as soon as one conditioning set has evidence below the threshold, since the minimum is then known not to cross.

Choosing the constants.

Validity does not depend on any of the following, and each is fixed before the monitored data arrive. The order k is the only choice with an assumption attached (Assumption 1): any k at least the maximum in-degree suffices, and a smaller k is valid as long as every non-adjacent pair has a separator of size at most k; otherwise the guarantee is lost, as the experiment on violations shows (Section 4.11). The prior scale g sets which effect sizes the test is tuned to detect: with variables standardised on the warm-up, a prior standard deviation of 0.5 ($g = 0.25$) concentrates power on coefficients of a few tenths, and Section 4.4 varies it. The split of α between families affects the thresholds only; the batch size affects how often the process is inspected and nothing else.

Properties of the output.

Because certificates are never retracted, the output is monotone: edges and arrowheads are added and never removed, and so the estimate at time t is contained in the estimate at every later time. Because the guarantee is time-uniform, the run may be stopped at any data-dependent time without changing the stated bounds, and the bound does not depend on the order in which pairs are examined. The guarantee concerns false assertions only. It says nothing about missed edges, and the number of undecided pairs is the price of that choice.

Relation to PC.

CERT-CD examines the same objects as the first layer of PC (partial correlations given small conditioning sets) but reverses the decision rule. PC removes (i, j) as soon as some S yields a non-rejection; CERT-CD adds (i, j) only when *every* S in the family yields rejection, which is why the family size multiplies computation but not the error bound: the bound for a non-adjacent pair uses only the one set that separates it.

A worked example.

Figure 1 follows the adjacency evidence for every pair in one random six-node DAG with Laplace errors (the first draw of a fixed seed, six true edges). The running maximum of $\log_{10} A_t$ is plotted against the number of observations. Evidence for the true edges rises past the threshold of $\lambda_A = \binom{6}{2}/0.05 = 300$ between 150 and 350 observations and is then frozen, since certified pairs are no longer updated; the evidence of

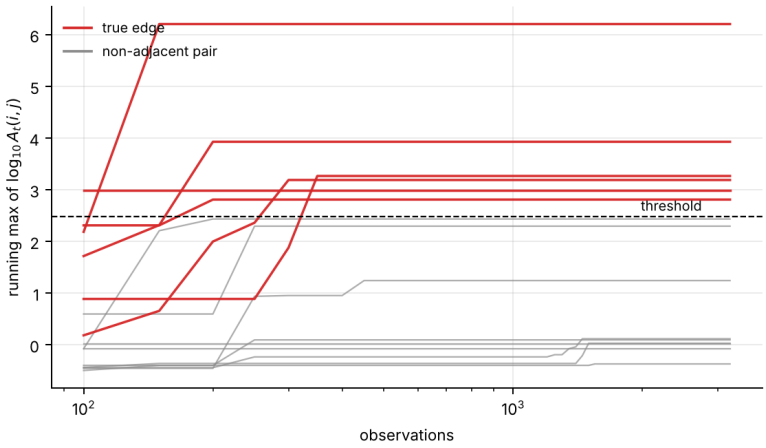


Fig. 1 Running maximum of the adjacency evidence for the 15 pairs of one six-node DAG with Laplace errors; $n = 3200$ observations. Red: true edges. Grey: non-adjacent pairs. Dashed: threshold.

the nine non-adjacent pairs stays below it. The highest of these, at about $10^{2.4}$, lies just under the threshold, which is the situation the family bound is designed to cover: with 15 pairs, a few non-adjacent pairs are expected to come close, and the threshold is placed so that, in the Gaussian model, none of them crosses with probability at least $1 - 2\alpha_A$.

4 Experiments

4.1 Protocol

The benchmarks were chosen to match the scope of the method: continuous data, linear or additive-noise mechanisms, a few hundred observations or more. The monitoring experiment (`exp13`) and the Tübingen evaluation (`exp14`) were committed to the repository with their protocols before they were run; the simulations of the pair orientation certificate (`exp16`) and of the primitive off the Gaussian model (`exp15`) were likewise written before running. Data outside the scope are reported in Section 4.11. Every number below comes from a JSON file in `results/`.

4.2 Monitoring: how often is a false edge ever reported?

We drew 150 random DAGs on $d = 8$ nodes (edge probability 0.25, coefficients with absolute value in $[0.5, 1.2]$) with Laplace and with Gaussian errors, and ran each method on nested prefixes of $n = 100, 200, 400, \ldots, 6400$ observations. A graph counts as a failure if any prefix's estimate contains an edge that is absent from the true skeleton, which is what an analyst who keeps every edge ever seen would report. CERT-CD ran as one stream with $\alpha = 0.1$ (so $\alpha_A = 0.05$ and a guaranteed bound of 0.1 in the Gaussian model), $k = 2$, warm-up of 50 rows and batches of 100. The comparators were PC (Fisher-z, $\alpha = 0.05$ and 0.01, `causal-learn` defaults), GES

Table 2 Probability that a false edge is ever reported over the grid $n = 100, \ldots, 6400$ in 150 random eight-node graphs (Wilson 95% interval for CERT-CD, and for the others where shown in the results file).

Method	Laplace errors	Gaussian errors
CERT-CD	0.027 $[0.010, 0.067]$	0.040 $[0.018, 0.085]$
PC, $\alpha = 0.05$	0.747	0.773
PC, $\alpha = 0.01$	0.267	0.320
GES-BIC	0.913	0.960
DirectLiNGAM	0.760	1.000

with the BIC score, and DirectLiNGAM with edges thresholded at $|B| > 0.05$, each rerun at every n.

CERT-CD reported a false edge in 4 of 150 graphs under Laplace errors and in 6 of 150 under Gaussian errors, against between 27% and 100% for the comparators (Table 2). The comparison is deliberately about monitoring: at a single sample size the comparators' false-edge rates are much lower, for example 42 of 150 graphs for PC at $\alpha = 0.05$ and $n = 6400$ under Laplace errors, so the gap is largely the cost of re-looking, which CERT-CD's guarantee absorbs. The price is coverage. At $n = 6400$ under Laplace errors CERT-CD's precision is 0.994 and its recall 0.938 ($F_1 = 0.965$), and it leaves 76.8% of the pairs undecided because it never certifies absence (Gaussian: 76.9%). By comparison, PC at $\alpha = 0.01$ has $F_1 = 0.970$ at the final sample size. CERT-CD's orientation is switched on by default here, which splits the budget; edges, not arrowheads, are scored.

The nonparametric certificate on the same data.

We reran the nonparametric primitive of Section 3.4.2 ($k = 2$, same level) on exactly the same 150 DAGs and prefixes (`exp25`). It reported a false edge at some sample size in 9 of 150 graphs under Laplace errors (0.060, Wilson $[0.032, 0.110]$) and 7 of 150 under Gaussian errors (0.047, $[0.023, 0.093]$), against 4 and 6 for the linear primitive. Both are consistent with the level $\alpha_A = 0.05$, and the asymptotic guarantee of the nonparametric version is visible as a small excess over the linear version, which is exact under Gaussian errors. Its skeleton precision and recall at $n = 6400$ are 0.988 and 0.933 under Laplace errors (0.990 and 0.932 under Gaussian), and it has no recall at $n = 100$ (the first rows are used to fit the regressions) and 0.05 at $n = 200$ under Laplace errors, against 0.21 and 0.65 for the linear primitive. For graph sizes $d = 4, 8, 12, 16$ (the scaling data below) it gives 0/60, 3/60, 2/60 and 5/60 graphs with a false edge, with precision ≥ 0.992 and recall between 0.91 and 0.95.

4.3 Sample size and graph size

Power by sample size.

The same runs give precision and recall of the skeleton at each prefix (Table 3, Laplace errors; Gaussian errors are similar). CERT-CD's precision is at least 0.99 at every n, but its recall is low where the sample is small: 0.21 at $n = 100$ and 0.65 at $n = 200$,

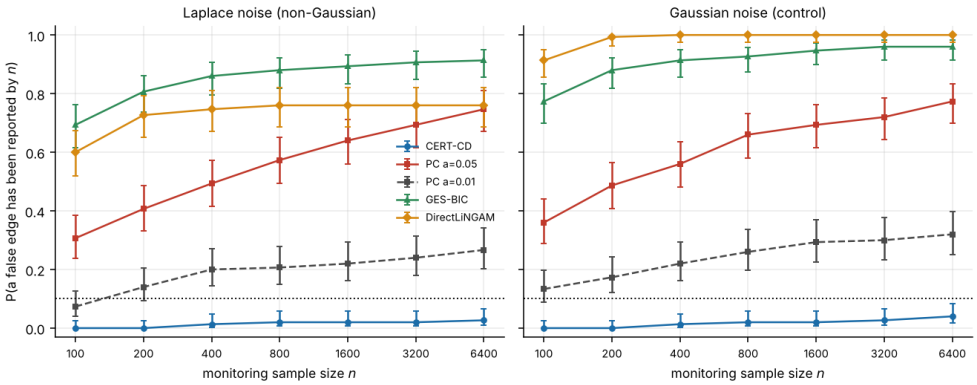


Fig. 2 Cumulative fraction of graphs with a false edge, by sample size, for each method (Laplace and Gaussian errors).

Table 3 Skeleton precision / recall by sample size, Laplace errors, 150 DAGs (`results/exp13_by_n.json`).

n	100	200	400	800	1600	6400
CERT-CD	1.00/0.21	1.00/0.65	1.00/0.82	1.00/0.87	0.99/0.91	0.99/0.94
PC $\alpha = 0.01$	0.99/0.78	0.98/0.85	0.98/0.89	0.99/0.92	0.99/0.94	0.99/0.96
PC $\alpha = 0.05$	0.94/0.83	0.95/0.87	0.96/0.90	0.96/0.93	0.96/0.94	0.95/0.96
GES-BIC	0.83/0.93	0.84/0.94	0.87/0.96	0.88/0.97	0.92/0.98	0.94/0.99
DirectLiNGAM	0.85/0.98	0.91/0.99	0.97/1.00	0.98/1.00	0.99/1.00	1.00/1.00

rising to 0.94 at $n = 6400$. The comparators recover most edges from $n = 100$ and pay with false edges. At a fixed large sample DirectLiNGAM is better than CERT-CD on this model (precision and recall 1.00 at $n = 6400$ against 0.99 and 0.94), because it uses the non-Gaussianity of every variable at once; the claim of CERT-CD is about monitoring, not about accuracy at a single sample size.

Graph size.

We repeated the monitoring experiment for $d = 4, 8, 12, 16$ with 60 DAGs per size, Laplace errors, expected degree about 1.75 and the grid $n = 100, \ldots, 3200$ (Table 4). The probability that CERT-CD ever reports a false edge is 0, 0.050, 0.067 and 0.083, rising with d but inside the guaranteed bound of 0.1 (which is exact only for Gaussian regression errors); the Wilson upper limit at $d = 16$ is 0.18, so 60 graphs cannot separate this from the bound. For PC at $\alpha = 0.05$, GES and DirectLiNGAM the probability rises to between 0.88 and 1.00 at $d = 12$ to 16, and for PC at $\alpha = 0.01$ to 0.67. CERT-CD's recall stays between 0.92 and 0.96, its precision above 0.99, and the fraction of pairs left undecided grows with d (0.56 at $d = 4$, 0.89 at $d = 16$), because the share of non-adjacent pairs grows and none of them is ever certified absent. The

Table 4 Monitoring by graph size (60 DAGs per size, Laplace errors, $n \leq 3200$): probability of ever reporting a false edge, with CERT-CD's Wilson interval (`results/exp17_summary.json`).

Method	$d = 4$	$d = 8$	$d = 12$	$d = 16$
CERT-CD	0.000 [0, 0.06]	0.050 [0.02, 0.14]	0.067 [0.03, 0.16]	0.083 [0.04, 0.18]
PC, $\alpha = 0.05$	0.233	0.667	0.883	1.000
PC, $\alpha = 0.01$	0.033	0.267	0.367	0.667
GES-BIC	0.317	0.933	1.000	1.000
DirectLiNGAM	0.217	0.867	0.967	1.000
CERT-CD recall at $n = 3200$	0.964	0.930	0.922	0.919

Table 5 Ablation, 20 DAGs, $d = 8$, Laplace errors, monitored to $n = 3200$. Default: $k = 2$, prior scale 0.5, $\alpha = 0.1$, orientation on. Precision and recall are of the skeleton at $n = 3200$ (`results/exp17_summary.json`).

Variant	ever false edge	precision	recall	median seconds
default	0/20	1.000	0.934	53
$k = 0$	20/20	0.526	1.000	145
$k = 1$	15/20	0.873	0.960	70
$k = 3$	0/20	1.000	0.921	53
no orientation	0/20	1.000	0.934	0.1
prior scale 0.25	0/20	1.000	0.934	52
prior scale 1.0	0/20	1.000	0.927	52
$\alpha = 0.05$	0/20	1.000	0.927	53
$\alpha = 0.2$	1/20	0.993	0.934	56

median runtime of one run at $n = 3200$ is 17, 45, 67 and 94 seconds, against under three seconds for each comparator.

4.4 Ablation of CERT-CD

We varied one constant at a time on 20 eight-node DAGs with Laplace errors ($n \leq 3200$, same monitoring criterion; Table 5).

The order k is the only constant that matters for validity. With $k = 0$ and $k = 1$, below the maximal in-degree of many of these graphs, the separator-size assumption is violated and false edges appear in every graph (20/20) and in 15/20 respectively, with precision 0.53 and 0.87; with $k = 3$ nothing changes relative to $k = 2$ except a slightly lower recall. The prior scale and the total level change recall by less than 0.01 and leave the number of graphs with a false edge at zero, except that $\alpha = 0.2$ gave one. Switching off orientation leaves the skeleton unchanged, as it should, since the adjacency certificates do not depend on it, and removes nearly all of the computing time: the adjacency certificates alone take about a tenth of a second per run, and the orientation processes dominate the cost.

Table 6 Nonlinear mechanisms, 60 DAGs per row, $d = 8$, $n \leq 3200$: graphs with a false edge at some sample size, and skeleton precision / recall at $n = 3200$.

		linear primitive		nonparametric	
Mechanisms	Errors	ever false	P / R	ever false	P / R
nonlinear	Laplace	33	0.83 / 0.66	1	0.997 / 0.83
nonlinear	Gaussian	28	0.85 / 0.62	2	0.995 / 0.97
linear	Laplace	3	0.99 / 0.93	4	0.99 / 0.92
linear	Gaussian	3	0.99 / 0.93	3	0.99 / 0.93

4.5 Nonlinear mechanisms

The linear primitive assumes linear mechanisms. We generated random DAGs ($d = 8$, edge probability 0.25, 60 per cell, $n \leq 3200$) in which every edge carries a nonlinear function drawn from $\{x, \tanh 2x, \sin 1.5x, x^2 - 1, |x| - 0.8\}$, with Laplace or Gaussian errors, and a linear control (`exp22`). The linear primitive reports a false edge at some sample size in 33 of 60 graphs under Laplace errors and 28 of 60 under Gaussian errors; the nonparametric primitive in 1 of 60 (Wilson $[0.003, 0.089]$) and 2 of 60 ($[0.009, 0.114]$), and it finds more of the true edges (recall 0.83 against 0.66 under Laplace errors, 0.97 against 0.62 under Gaussian errors; precision 0.997 and 0.995 against 0.83 and 0.85). In the linear control the two primitives agree (false edges in 3 and 4 of 60 graphs under Laplace errors, 3 and 3 under Gaussian errors; recall 0.93 against 0.92).

4.6 The adjacency primitive: cost of anytime validity and calibration off-model

Cost and benefit of the sequential test.

With 2000 replications under the independence null and monitoring up to $n = 1000$, a Fisher-z test inspected at every sample size falsely rejects in 34.6% of runs at nominal $\alpha = 0.05$ (56.5% at 0.10, 81.6% at 0.20; 10.0% at 0.01). The same test at one fixed n rejects 5.6% (0.01: 1.0%), and the e-process 2.4% (0.01: 0.7%). The e-process is conservative by about a factor of two in this setting, which is the price of the time-uniform guarantee.

Off the Gaussian model.

The primitive is exact only for Gaussian regression errors. Table 7 reports the realised probability of ever crossing $1/\alpha$ over $n = 50, \ldots, 1000$ (monitored at every observation), with 1000 replications per cell, for a null pair with no covariate effect (S0), a chain $i \leftarrow k \leftarrow j$ tested given k, and a heteroscedastic null, under four error laws. A robust t-statistic confidence sequence (Lindon et al., 2026) is shown for comparison. For Gaussian and Laplace errors without heteroscedasticity the primitive stays below nominal. For skewed or heavy-tailed errors it is up to 2.5 times nominal at $\alpha = 0.01$ (0.025 for lognormal errors), and under heteroscedastic errors it exceeds nominal at

Table 7 Realised probability of ever crossing, $n = 50, \ldots, 1000$ monitored every observation (primitive, $g = 0.25$) versus a robust-t sequence checked every five observations, 1000 replications per cell.

Scenario	Errors	BF .05	BF .01	robust-t .05	robust-t .01
S0	Gaussian	0.023	0.006	0.044	0.009
S0	Laplace	0.016	0.004	0.032	0.009
S0	$t_{2.5}$	0.035	0.019	0.067	0.021
S0	lognormal	0.043	0.025	0.098	0.035
Chain	Gaussian	0.021	0.008	0.042	0.012
Chain	Laplace	0.018	0.006	0.040	0.011
Chain	$t_{2.5}$	0.026	0.013	0.076	0.035
Chain	lognormal	0.039	0.019	0.086	0.026
Het.	Gaussian	0.069	0.025	0.066	0.022
Het.	Laplace	0.065	0.028	0.053	0.017
Het.	$t_{2.5}$	0.042	0.025	0.077	0.035
Het.	lognormal	0.061	0.032	0.089	0.037

$\alpha = 0.05$ even for Gaussian errors (0.069). The robust version is not uniformly better: it is closer to nominal under heteroscedasticity but wider of the mark in the heavy-tailed cells (0.098 at $\alpha = 0.05$ for lognormal errors in S0). In the six-noise-law calibration of the earlier experiment (`exp8`, 2000 replications, $n \leq 800$) every cell is below nominal, the largest realised-to-nominal ratio being 0.95 and the median about 0.48. We therefore treat the adjacency bound as exact in the Gaussian model and as an approximation otherwise; in the stated experiments the approximation is worse under heteroscedasticity and heavy tails than the earlier calibration suggested.

4.7 Graphs that violate strong faithfulness

We generated 1780 random linear-Gaussian DAGs, of which 56.2% satisfy the separator-size premise, and stratified them by whether δ-strong faithfulness ($\delta = 0.15$) holds (1000 graphs) or fails (500 graphs). In the strata where it holds CERT-CD certified no false edge (0 of 1000) and made 8 wrong-arrow certifications (rate 0.008); where it fails it certified 6 false edges in 500 graphs (0.012, Wilson $[0.006, 0.026]$) and 29 wrong arrows (0.058, $[0.041, 0.082]$; one-sided binomial test against 0.05, $p = 0.23$). PC lost a true edge in 8 of 1000 and 241 of 500 graphs (0.482 $[0.439, 0.526]$) and our earlier delete-based algorithm, SPRINT-CD, in 0 and 169 (0.338 $[0.298, 0.381]$). These are different error types and are not commensurable: a lost edge is an omission, a false edge a commission, and CERT-CD pays for its few commissions with undecided pairs (75% undecided in this experiment; 97% of true edges certified and 94% oriented, against a CPDAG ceiling of 28% of edges orientable).

Table 8 Pair orientation, 50 simulated pairs per family, $n = 800$. Entries are (decided, wrong). Forced is the accuracy when a direction is always output.

Family	agree	gg	ggh	forced gg	forced ggh
linear, Laplace	(3,0)	(27,0)	(3,0)	0.96	0.46
linear, uniform	(0,0)	(49,0)	(7,7)	1.00	0.34
linear, Gaussian	(0,0)	(0,0)	(0,0)	0.44	0.44
ANM, tanh	(41,0)	(46,0)	(44,0)	1.00	1.00
ANM, square	(18,0)	(46,0)	(18,0)	1.00	1.00
ANM, sine	(43,0)	(50,0)	(43,0)	1.00	1.00
heteroscedastic Gaussian	(0,0)	(32,32)	(4,0)	0.06	1.00
strongly heteroscedastic	(0,0)	(9,9)	(25,0)	0.64	0.98
heteroscedastic Laplace	(39,0)	(45,0)	(40,0)	1.00	1.00

4.8 Orientation of pairs

Design families.

The nine families of Table 8 were used while the certificate was being designed: the agreement rule was adopted because each class alone failed on some of them, so the absence of wrong certifications there is not independent evidence. We simulated 50 pairs of $n = 800$ per family and ran the certificate at $\alpha = 0.05$. The agreement certificate committed to a wrong direction in none of the 450 pairs. The constant-scale class certifies the wrong direction on all 32 and 9 pairs it decides in the two heteroscedastic Gaussian families, and the location-scale class is wrong on all 7 it decides for uniform linear errors. Agreement commits on all the additive-noise families (for example 41 of 50 for the tanh mechanism) and on Laplace heteroscedastic pairs (39 of 50), declines the strongly heteroscedastic Gaussian pairs, and declines linear Gaussian pairs, which are not identifiable.

Held-out families.

To measure behaviour outside the design set we fixed nine further families before running (`exp18`), none used during design, with 30 pairs of $n = 800$ each (Table 9), and compared with four forced-choice published rules (IGCI, RECI, ANM-HSIC and a pairwise LiNGAM rule; our re-implementations, Section 4.1).

The agreement certificate made no wrong certification in eight of the nine families, and wrong certifications on all 5 pairs it decided in the family with a bimodal cause and linear Gaussian noise (5 of 30 pairs, so the overall error rate is 5 of 270 pairs, or 5 of the 83 pairs that it decided). In that family a generalised-Gaussian density cannot represent the mixture marginal of the cause, so the factorisation with the bimodal variable as the effect is fitted better; this is a misspecification of the model class, not a violation of validity, and both classes commit the same error. The agreement certificate commits less often than either class alone (83 of 270 pairs), and the classes alone err in two further families (the constant-scale class on post-nonlinear pairs, the location-scale class on skewed linear errors). Of the forced-choice rules, ANM-HSIC is perfect on seven families and completely wrong on the two with multiplicative and

Table 9 Held-out families, 30 pairs of $n = 800$ each. For the certificates: decided / wrong. Forced-choice columns: accuracy, always outputting a direction.

Family	agree	gg	ggh	IGCI	RECI	ANM	LiNGAM
cubic ANM	6/0	14/0	6/0	1.00	1.00	1.00	0.07
exponential ANM	28/0	30/0	28/0	1.00	1.00	1.00	1.00
logistic ANM	21/0	27/0	21/0	0.37	0.50	1.00	0.00
linear, t_3 errors	0/0	8/0	0/0	0.43	0.37	1.00	1.00
linear, skewed errors	0/0	17/0	12/12	0.10	0.37	1.00	1.00
multiplicative noise	6/0	7/0	17/0	1.00	1.00	0.00	0.53
heteroscedastic, cosine	17/0	26/0	20/0	1.00	1.00	1.00	0.23
post-nonlinear	0/0	4/4	14/0	1.00	0.97	0.00	0.87
bimodal cause, linear	5/5	6/6	10/10	0.40	0.83	1.00	1.00

Table 10 Agreement certificate by sample size (decided, of 20 pairs; wrong in parentheses); $n = 800$ has 30 pairs. Also the constant-scale class alone for skewed linear errors.

Family	$n = 200$	400	800 (30)	1600
cubic ANM	0 (0)	0 (0)	6 (0)	15 (0)
heteroscedastic, cosine	3 (0)	6 (0)	17 (0)	19 (0)
linear, skewed errors	0 (0)	0 (0)	0 (0)	0 (0)
constant-scale class alone	0 (0)	1 (0)	17 (0)	18 (0)

post-nonlinear structure; the others are perfect on some families and at or below chance on others, with no rule dominating. The certificate trades coverage for a lower error rate on this set, and it does not rule out failures in model classes it cannot represent.

Sample size.

For three of the held-out families we varied n (Table 10; 20 pairs per cell, and the $n = 800$ column from Table 9 with 30). The agreement certificate commits more often as n grows and made no wrong certification in any of these cells. The cubic and cosine families go from no or few commitments at $n = 200$ to 15 and 19 of 20 at $n = 1600$. For skewed linear errors it never commits: the constant-scale class alone is right on 18 of 20 at $n = 1600$, but the location-scale class certifies the wrong direction on 15 of 20, so the agreement rule stays silent, which is the protection it is meant to give and also its cost. The four forced-choice rules have their own regimes in these families: ANM-HSIC is right throughout, and the LiNGAM rule fails on the cubic and cosine families.

4.9 The Tübingen cause-effect pairs

The protocol was fixed before the run: all 102 scalar-scalar pairs, official weights, tied values dequantised, a random permutation, at most 2000 rows per pair, $\alpha = 0.05$,

Table 11 Tübingen pairs (80 scored of 102). Accuracy is among decided pairs; interval is Clopper–Pearson 95%; p is the one-sided binomial test of accuracy $> 1/2$.

Certificate	decided	correct	accuracy	95% interval	p
agree (primary)	12	11	0.917	[0.615, 0.998]	0.003
gg only	23	19	0.826	[0.612, 0.950]	0.001
ggh only	18	14	0.778	[0.524, 0.936]	0.015

Table 12 Tübingen pairs: forced-choice accuracy (unweighted / weighted by the official weights) on the 80 scored pairs and, for the baselines, on all 102.

Rule	80 scored pairs	all 102 pairs
CERT-CD constant scale (forced)	0.725 / 0.65	–
CERT-CD location-scale (forced)	0.700 / 0.705	–
IGCI	0.537 / 0.689	0.559 / 0.664
RECI	0.662 / 0.758	0.608 / 0.709
ANM-HSIC	0.650 / 0.618	0.569 / 0.585
pairwise LiNGAM	0.562 / 0.565	0.608 / 0.620
CERT-CD agreement, decided only	11 of 12 correct (0.917)	–

seed 20261023. Pairs with fewer than 100 post-warm-up rows (22 of 102) cannot be scored and count as abstentions; 80 pairs are scored. This protocol was written after an earlier linear certificate decided 14 pairs and was right on 6; no parameter of the relative-fit certificate was tuned on these data.

The agreement certificate decided 12 of 80 scored pairs (15%, which is 11.8% of all 102) and was right on 11 (Table 11); weighted by the official weights its accuracy is 0.816. Forced to always output a direction, our two classes are right on 72.5% (constant scale, 0.65 weighted) and 70% (location scale, 0.705 weighted) of the 80 scored pairs. Table 12 compares them with the four forced-choice rules computed under the same preprocessing. On the 80 scored pairs the best rule by unweighted accuracy is RECI (66.2%), and RECI is also the best by weighted accuracy (0.758, against 0.65 and 0.705 for our classes). So the forced predictions of our forecasters are not uniformly better than a simple regression-error rule; the contribution of the certificate is the abstention, which turns a 70–72% forced accuracy on this set into 11 of 12 on the pairs it commits to. Published methods report forced accuracies of roughly 63–83% on all pairs (Immer et al., 2023; Mooij et al., 2016; Tagasovska et al., 2020) under their own protocols, and abstaining methods in the literature report 75–85% at coverages of 35–85%; these are not comparable to ours, which are on a subset, with abstention, at a pre-fixed α. The evidence is modest: 12 decisions, a wide interval, and a statement about predictive fit.

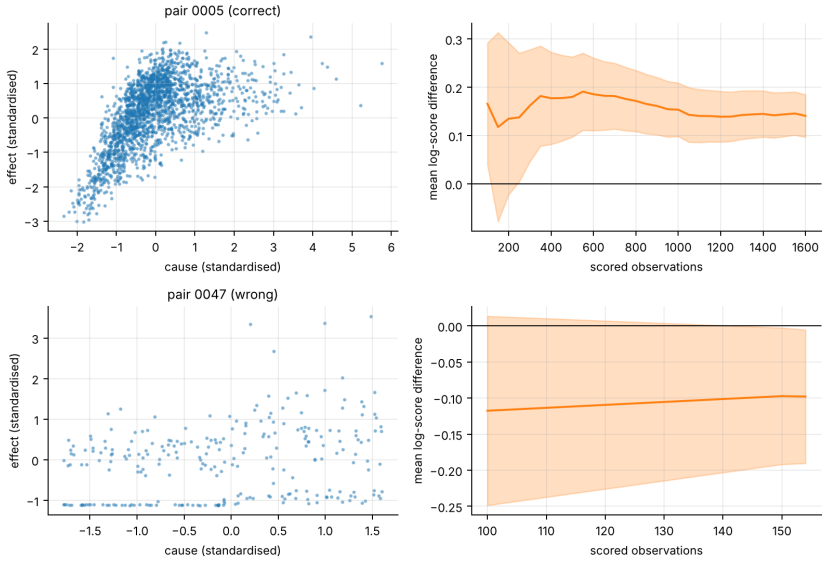


Fig. 3 Tübingen pairs 5 (top) and 47 (bottom): scatter plot and confidence sequence for the running mean log-score difference under the constant-scale class; a lower end above zero certifies the true direction.

Sensitivity to the random draws.

The protocol uses one seed for the row permutation, the subsample of long pairs and the dequantisation of ties. We repeated it with three further seeds (`exp19`), keeping everything else fixed. The agreement certificate decided 9, 11 and 10 pairs and was right on 9, 11 and 10 respectively, against 11 of 12 for the frozen seed; the constant-scale class decided 17 to 24 pairs with 14 to 20 correct. The counts are of the same pairs under different random draws and are therefore not independent evidence, but they show that the headline is not an artefact of one lucky permutation. The one error of the frozen seed (pair 47) did not recur.

Figure 3 shows two pairs: pair 5, the first on which the agreement certificate commits, correctly, and pair 47, the only pair on which it commits and is wrong; the wrong decision is made on 254 observations, with 154 scored, and the confidence sequence only just excludes zero.

4.10 Real physical data with a known graph: the Causal Chambers

The Causal Chambers (Gamella, Peters, & Bühlmann, 2025) are two physical devices, a wind tunnel and a light tunnel, whose causal graphs are known from the construction of the devices and which provide real measurements. We used two recorded experiments in which the actuators are randomised, fixed before any result was seen (`exp20`): the wind tunnel with random intake loads (`random_loads_intake`, 10 varying variables, 8 true edges among them) and the light tunnel with uniform inputs

(`uniform_reference`, 20 variables, 39 true edges). A third experiment, the wind tunnel with random loads on both fans, was dropped before running because the two load variables are identical in the recording, so the two nodes cannot be told apart. Variables are the nodes of the chamber graph with at least 20 distinct values; no node is hidden. For each dataset we drew 10 random subsamples of 4000 rows (rows are shuffled, to break the serial dependence of the recording) and ran each method on nested prefixes $n = 250, \ldots, 4000$, as in Section 4.2.

First result: every method reports false edges.

With the linear certificate at $k = 2$, and with all four comparators, a false edge (an edge of the algorithm's output that is not an edge of the chamber graph) was reported in every one of the 10 subsamples of both devices (rows "strict" of Table 13). CERT-CD's precision at $n = 4000$ was 0.40 and 0.41, similar to PC's, and DirectLiNGAM was the most accurate on the light tunnel (0.79 precision, 0.59 recall). This was the result that motivated the changes below, so what follows is a redesign after seeing these data, not a test on fresh ones.

Why: two violated assumptions and a graph that is not Markov to the data.

Inspecting the persistent false edges showed three things. First, in the light tunnel the separator of ir_1 from ir_2 is the five parents of ir_1, larger than $k = 2$, so Assumption 1 fails; raising k to 3 removes all but three pairs. Second, the mechanisms are nonlinear, which the nonparametric primitive of Section 3.4.2 handles. Third, some pairs the graph calls non-adjacent are dependent given the graph's own separating set. To make this precise without using any of the algorithms under test, we computed, for every non-adjacent pair, the generalised covariance statistic given the parents of the non-descendant (additive spline regression) on 20000 rows, and call the pairs with $|z| > 5$ the *reference set* V (`exp23`). It has 11 pairs in the wind tunnel (the four pressure sensors are mutually dependent given their parents, with residual correlation 0.999, and the two rotation speeds are dependent) and 5 in the light tunnel (the infrared and visible sensors of the same unit, residual correlation 0.75–0.80). These are dependences that the graph cannot explain, plausibly a shared source or drift that the graph does not contain. A certified edge that is neither a graph edge nor in V is called a *false edge, adjusted*; the scoring is partly circular for any method that detects conditional dependence with $n = 20000$, and we report both versions.

Result after the redesign.

With the nonparametric primitive and $k = 3$, the adjusted count is 1 of 10 subsamples in the wind tunnel (Wilson $[0.018, 0.404]$, 0.1 false edges per subsample) and 0 of 10 in the light tunnel ($[0, 0.278]$), where every other configuration in the table is at 10 of 10, except the linear certificate at $k = 3$ with 4 of 10. The nonparametric certificate certifies 10.1 edges per subsample in the light tunnel, of which about 7 are graph edges and 3 are pairs of V, and its recall of the graph edges is only 0.12 and 0.18. The strict count remains 10 of 10 because the pairs of V are certified, which is the intended behaviour for pairs whose dependence the graph does not explain, but it

Table 13 Causal Chambers, 10 subsamples of 4000 rows each. "Adjusted": graphs with a false edge at some sample size (of 10) and false edges per subsample at $n = 4000$, counting a certified pair as correct if it is a graph edge or in the reference set V; "strict": false edges per subsample counting graph edges only. Recall is of the graph edges (8 and 39). Wind tunnel: $d = 10$, $|V| = 11$; light tunnel: $d = 20$, $|V| = 5$. The CERT-CD variants have orientation off.

	wind tunnel				light tunnel			
Method	adj. ever	adj./run	strict/run	recall	adj. ever	adj./run	strict/run	recall
CERT-CD nonparametric, $k = 3$	1	0.1	5.9	0.12	0	0.0	3.0	0.18
CERT-CD linear, $k = 3$	10	1.1	3.7	0.39	4	0.4	3.5	0.18
CERT-CD nonparametric, $k = 2$	1	0.2	6.2	0.12	10	5.6	9.6	0.19
CERT-CD linear, $k = 2$	10	1.1	4.6	0.39	10	5.9	9.9	0.18
PC, $\alpha = 0.05$	10	2.0	7.9	0.53	10	7.9	11.9	0.24
PC, $\alpha = 0.01$	10	1.8	7.2	0.44	10	6.4	10.4	0.22
GES-BIC	10	8.3	16.8	0.69	10	8.9	12.9	0.60
DirectLiNGAM	10	4.6	10.8	0.61	10	2.9	6.1	0.59

means that against the graph alone CERT-CD still reports false edges. The baselines are scored on the same subsamples; PC, GES and DirectLiNGAM were run with their usual settings and were not given a larger conditioning order or a nonparametric test.

Direction.

The orientation certificate does much better. Among the edges certified by CERT-CD that are true edges, the bivariate agreement certificate decided 30 of 31 in the wind tunnel and 27 of 70 in the light tunnel (the constant-scale class alone 50 of 70), and every decision was correct (30/30 and 27/27). Separately, we applied the certificate to every true directed edge among the varying variables, each on two random subsamples of 2000 rows (`exp21`, 16 and 78 pair-runs for the two devices). The agreement certificate committed on 8 of 16 and 23 of 78 and was right every time (31/31); the constant-scale class alone committed on 8 and 40, and the location-scale class on 9 and 25, also always correctly. The forced accuracy of the two classes is 0.69 and 0.69 in the wind tunnel and 0.74 and 0.81 in the light tunnel, against 0.88 and 1.00 for RECI, 1.00 and 0.92 for IGCI, 0.38 and 0.81 for ANM-HSIC and 0.81 and 0.77 for the pairwise LiNGAM rule. As on the Tübingen pairs, the forced accuracy of the forecasters is not better than a simple rule, and the certificate is valuable for what it declines to decide. The causes in these experiments are uniformly distributed actuators, an easy case for identifiability; the 31 decisions are 47 edges analysed twice and not independent.

4.11 Scope: where the certificates fail

Assumption violations.

We simulated 200 datasets of $n = 3000$ for each of four violations at $\alpha = 0.05$. In-model there were no false edges and no wrong arrows. Violating the separator-size premise gave a false edge in 200/200 runs and false arrows in 26; a latent common cause gave a certified edge in 200/200 runs, which is a MAG adjacency and not an

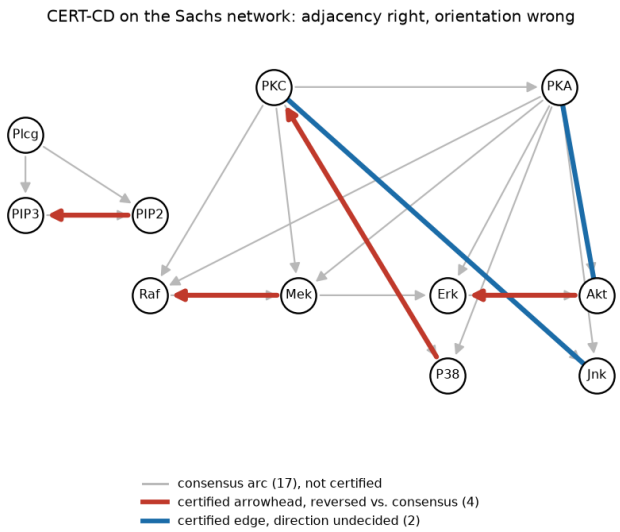


Fig. 4 Sachs data: certified edges and arrows against the consensus network.

error in that sense, with 38 wrong arrows; nonlinear mechanisms gave false edges in 200/200 runs and 142 wrong arrows; Poisson counts gave false edges in 200/200 runs and 28 wrong arrows. The adjacency certificate is valid only for the linear model with independent errors, and the orientation certificate inside the multivariate algorithm fails badly under nonlinearity.

Sachs protein-signalling data.

The data (Sachs, Perez, Pe'er, Lauffenburger, & Nolan, 2005) are discrete-like, nonlinear and heavy-tailed and sit outside the method's scope, which is why we report them here. On 853 cells (753 after a 100-cell warm-up) CERT-CD certified 6 edges among 55 pairs, all in the consensus skeleton, left 49 pairs undecided, and certified 4 arrowheads, all reversed relative to the consensus network. PC found 7 edges, all in the consensus skeleton. DirectLiNGAM's causal order contradicts the consensus direction on all six pairs CERT-CD certified. Consensus directions are themselves contested (Mooij, Magliacane, & Claassen, 2020), but the certified arrows are confidently unreliable, and nothing in the output says so. A comparison with the iterative protocol of Uehara (2026) (precision 0.933, recall 0.700 with 21 queries) is not like for like: the protocol, the ground truth and the interaction model differ.

5 Discussion

What is established.

Under monitoring, the adjacency certificate kept the probability of ever reporting a false edge near 3–4% in random eight-node graphs, where the comparators reached

27% to 100%, and its stated bound is exact in the Gaussian regression model. The orientation certificate commits rarely and, on the Tübingen pairs under a frozen protocol, was right on 11 of the 12 pairs on which it committed.

Limitations.

- *The adjacency bound is asymptotic off the Gaussian model.* The null that a partial regression coefficient is zero follows from d-separation in a linear model with independent errors, but the primitive is exact only for Gaussian regression errors. Section 4.6 measures how far from nominal it is under skewed, heavy-tailed and heteroscedastic errors; we have no finite-sample correction. Conditional independence testing without a model restriction is impossible (Shah & Peters, 2020), so none is to be expected without further assumptions.
- *Power at small samples.* Skeleton recall is 0.21 at $n = 100$ and 0.65 at $n = 200$, and at a single large sample size DirectLiNGAM is more accurate on the same data. The case for CERT-CD is monitoring, not single-sample accuracy.
- *The order k.* Validity needs k at least the size of a separator for every non-adjacent pair; $k = 0$ and $k = 1$ produced false edges in most graphs (Section 4.4), and nothing in the output warns of it.
- *Real physical devices.* On the Causal Chambers every method, including the first version of CERT-CD, reported edges that the graph does not contain in every subsample, with precision around 0.4. The redesign (nonparametric test, $k = 3$) was made after seeing these data, so its result there is not an independent test; it reports no edge outside the graph and the reference set of dependences the graph cannot explain in 9 and 10 of 10 subsamples, but its recall of the graph edges is 0.12 and 0.18 (Section 4.10). The false-edge guarantee should not be read as a property of any real dataset, and it needs k at least the size of the largest separator, which is not known in practice.
- *Model-class errors in orientation.* On a family with a bimodal cause and Gaussian noise, the agreement certificate was wrong on all 5 pairs it decided, because a generalised-Gaussian marginal cannot represent the cause.
- *Most pairs stay undecided.* About three quarters of the pairs in the eight-node experiments are undecided at $n = 6400$ because the algorithm never certifies absence, and the orientation certificate decided 15% of the scorable Tübingen pairs. Certificate-only output trades coverage for validity, and a user who needs a complete graph needs a different method.
- *Predictive fit is not causation.* The orientation certificate certifies a better expected log score. It coincides with the causal direction under additive-noise or linear non-Gaussian identifiability and under causal sufficiency; under confounding it need not, and the multivariate algorithm assumes causal sufficiency.
- *The orientation layer inside the multivariate algorithm is the old linear certificate.* Its guarantee is conditional on a linear generalised-Gaussian model and fails outside it (Section 4.11). Conditioning the relative-fit forecasters on the frozen block would remove this, and is not done here.

- *Assumed, not checked.* The linear structural model with independent errors is an assumption. Residual diagnostics on the warm-up prefix can flag gross departures, but they are not part of the guarantee.
- *Small evidence base for orientation.* The Tübingen result rests on 12 committed pairs out of 80 scorable, and 22 of the 102 pairs were too short or discrete to score; the confidence interval on the accuracy is wide. The method was designed after an earlier version failed on the same data, without tuning on it, and Section 4.9 reports the protocol that was fixed.
- *Domain.* The experiments use continuous data from mostly low-dimensional systems. The Sachs data, which are not of this kind, are reported as out-of-scope behaviour (Section 4.11).

6 Conclusion

Asserting a feature of a causal graph only by rejecting a null that is true when the feature is absent removes the dependence of the false-edge guarantee on faithfulness and makes the guarantee survive monitoring. In simulation, this holds at a few percent where deletion-based methods reach most of the graphs, with the cost that most pairs are left undecided. A predictive-fit orientation certificate extends the same discipline to direction and committed rarely but mostly correctly on a standard benchmark. Both certificates are valid only within their models, and the paper reports where they fail.

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Ethics approval. Not applicable. The study involves no human participants or animals. The protein-signalling data analysed in Section 4.11 are publicly available and were published by Sachs et al. (2005).

Data and code availability. All simulation code, experiment scripts, random seeds and figure-generating scripts are available at https://github.com/vinhqdang/SPRINT-CD. The protein-signalling data of Sachs et al. (2005) are redistributed with the `bnlearn` package (Scutari, 2010) and are downloaded by the experiment script rather than included in the repository. The Tübingen cause-effect pairs of Section 4.9 are downloaded on first use from https://webdav.tuebingen.mpg.de/cause-effect/ and are not redistributed.

Author contributions. The author is solely responsible for all aspects of this work.

Use of generative AI. A large language model was used in support of writing this manuscript. The author verified and modified the resulting content and takes full responsibility for the manuscript in its entirety, including its claims, derivations, experiments and references.

Appendix A The safe linear primitive

Setup.

Let $y \in \mathbb{R}^n$ stack the observations of the response X_i, x those of the tested covariate X_j, and $W \in \mathbb{R}^{n\times q}$ the conditioning block with an intercept, $q = |S| + 1$. The model is $y = W\alpha + \beta x + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$, with null $\beta = 0$. Let $M = I - W(W^\top W)^{-1}W^\top$ and $\tilde{y} = My$, $\tilde{x} = Mx$, and write $S_{ij} = \tilde{x}^\top\tilde{y}$, $S_{ii} = \tilde{y}^\top\tilde{y}$, $S_{jj} = \tilde{x}^\top\tilde{x}$, so that $\hat{r}^2 = S_{ij}^2/(S_{ii}S_{jj})$.

Marginal likelihoods.

The null is invariant under $y \mapsto \lambda y + Wc$, $\lambda > 0$, and the right-Haar prior is $\pi(\alpha, \sigma) \propto \sigma^{-1}$. Integrating α and σ gives $m_0(y) = C_{n,q} S_{ii}^{-(n-q)/2}$. Under the alternative, put $\beta \mid \sigma \sim N(0, g\sigma^2)$. Completing the square in β,

$$m_1(y) = C_{n,q}(1 + gS_{jj})^{-1/2}\Big(S_{ii} - \frac{S_{ij}^2}{S_{jj} + g^{-1}}\Big)^{-(n-q)/2},$$

and with $G = gS_{jj}$, $\kappa = G/(1+G)$,

$$E_n = \frac{m_1}{m_0} = (1+G)^{-1/2}\big(1 - \kappa\hat{r}_n^2\big)^{-(n-q)/2},$$

which is (3). The constant g is fixed before the monitored data arrive, from a discarded warm-up prefix.

Proof of Proposition 4.

A Bayes factor with a right-Haar prior on a group acting on the nuisance parameters of both hypotheses equals the Bayes factor of the maximal invariant (Berger, Pericchi, & Varshavsky, 1998; Hendriksen et al., 2021). The maximal invariant's law under the null does not depend on (α, σ), so E_n is a ratio of a density to its null density in the filtration $(\mathcal{G}_n)$ generated by the maximal invariants, hence a test martingale there. Ville's inequality gives $\mathbb{P}(\exists n : E_n \geq 1/\alpha) \leq \alpha$ under every Gaussian null.

The raw-observation filtration.

(E_n) is not a martingale in the filtration of the raw observations. Take $S = \emptyset$, $g = 0.25$ and $X_{1:2} = Y_{1:2} = (-1, 1)$, with the Gram matrix centred. Then $E_2 = 1$, while under an i.i.d. standard Gaussian independence null the exact conditional expectation is $\mathbb{E}[E_3 \mid \mathcal{F}_2] \approx 1.083 > 1$. The guarantee is therefore the Ville bound on the crossing event, which is what Propositions 3 and Theorem 7 use; it is not a claim about $\mathbb{E}[E_\tau]$ at arbitrary stopping times of the raw filtration.

Appendix B The linear orientation certificate inside the multivariate algorithm

Inside the streaming algorithm, direction is certified after the pair's adjacency, with a sequential universal-inference ratio (Wasserman, Ramdas, & Balakrishnan, 2020). For a pair (i, j) certified at the stopping time T, let Z_{ij} be a block of conditioning variables fixed at T. The numerator is a generalised-Gaussian linear factorisation $x \to y$ given Z_{ij} fitted sequentially on the past after T, with shape in $[0.9, 8]$ and a continuation across the Gaussian boundary of the shape grid; the denominator is the supremum over the reversed linear factorisation. The observations before T are discarded. Conditionally on $\mathcal{F}_T$ the later observations are i.i.d. and Z_{ij} is a fixed set, so the usual universal-inference argument gives an e-process for the null that the reversed linear factorisation holds; the union bound over the $d(d-1)$ ordered pairs uses the fixed family and does not depend on which hypotheses the run examines. The guarantee is conditional on the linear generalised-Gaussian model describing the pair given Z_{ij}, and it fails outside it (Section 4.11). A direction is reported after at least 80 post-certification observations.

Appendix C Inverting the primitive into a confidence sequence

Replacing y by $y - \beta_0 x$ gives the process for the null $\beta = \beta_0$, and the set of β_0 not rejected at $1/\alpha$ is the interval

$$\hat{\beta} \pm \frac{1}{S_{jj}} \sqrt{\frac{\tau(S_{ii}S_{jj} - S_{ij}^2)}{1-\tau}}, \qquad \tau = \frac{1}{\kappa}\Big[1 - \Big(\frac{\alpha}{\sqrt{1+G}}\Big)^{2/(n-q)}\Big],$$

valid for $0 < \tau < 1$. Since $\alpha < 1$, the term in brackets is positive and $\tau > 0$ for every $n > q$; when $\tau \geq 1$ the set is the whole line. The closed form is checked in the test suite against a numerical inversion over a grid of β_0.

Appendix D Additional experimental detail

Hyperparameters.

Warm-up of 50 discarded rows for the adjacency stream; prior standard deviation 0.5 on the standardised scale ($g = 0.25$); $k = 2$; adjacency share $\alpha_A/\alpha = 0.5$; batch size 250 in the graph experiments. For the pair orientation certificate: $\alpha = 0.05$, ρ tuned to $t^* = 200$, a 60-observation minimum, warm-up $\max(100, \lceil 0.2n \rceil)$.

Reproduction.

`python experiments/run_all.py` reproduces the adjacency experiments. The scripts `exp13_monitoring_fwer.py`, `exp14_tuebingen_relfit.py`, `exp15_primitive_offmodel.py` and `exp16_pair_orientation_sim.py` reproduce the tables of Section 4; seeds are fixed in each script and the results are written to `results/` as JSON.

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